

Research article

Spatio-temporal evolution and influencing factors of high quality economic development: Case study of Guangdong-Hong Kong-Macao Greater Bay Area

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ABSTRACT

With the increase in global economic integration, high-quality economic development (HQED) has become a common goal of all countries. Based on these five development concepts, this paper uses the Gini coefficient, trend surface analysis, geographically weighted regression (GWR), the entropy weighting method, and standard deviation ellipse analysis to study the spatio-temporal pattern and driving mechanism of HQED in the Guangdong-Hong Kong-Macao Greater Bay Area (GBA). This paper examines the spatial and temporal patterns and driving mechanisms of HQED in the Guangdong-Hong Kong-Macao Greater Bay Area (GBA) based on five development concepts. The study revealed that (1) HQED is on the rise overall, but at the same time, it highlights the uneven development of multiple dimensions, especially in terms of significant differences in innovation, openness, and sharing. (2) HQED shows a clear center-periphery structure, with Guangzhou, Shenzhen, and Hong Kong as the core high-value areas; the growth rate of HQED capacity in the internal areas is significantly greater than that in the external areas, and HQED is prominent in the cities around Guangzhou, Shenzhen, and Hong Kong. (3) Factors such as agglomeration level, human capital, foreign investment, infrastructure development, financial and environmental protection expenditures, and financial inputs, and scientific, and technological inputs have a significant positive effect on HQED, and their interactions are further strengthened. This study reveals the importance of the realization of HQED in the GBA and the promotion of the overall development of the region.

1. Introduction

China's economy has made significant progress since the inception of the country's reform and opening-up policy. In 2020, the total volume of China's economy exceeded 100 trillion yuan, and its per capita GDP exceeded \$10,000 for two consecutive years. As the global economy continues to weaken, trade protectionism is on the rise, and international competition has become more intense. China, on the other hand, is facing problems such as insufficient independent innovation capacity, the disappearance of its labor cost

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advantage, and a slowdown in economic development. The conventional primitive form of development is characterized by “high levels of investment, consumption, and pollution.” is gradually not applicable to the current economic growth requirements. [1]. The Chinese government has proposed a new strategy of HQED to respond to the new normal economic development and address the inevitable requirements of social conflicts [2]. The 20th Party Congress re-emphasized the need to focus on promoting HQED and driving the economy toward achieving effective qualitative enhancement and balanced quantitative growth. Throughout the world’s scientific and technological innovation and HQED, the GBA has become a symbol of leading the world in HQED. As China’s frontier pilot for modernization, the GBA is the fourth-largest bay area in the world alongside the San Francisco Bay Area, the New York Bay Area, and the Tokyo Bay Area [3]. The GBA is the Bay Area with the shortest time of establishment, the largest area, and the fastest rate of GDP growth, has an extremely broad space for growth, is the front-runner in fostering sustained competitiveness and creativity, and is the leader in promoting domestic and international double-cycle strategies at home and abroad. With the arrival of the fifth year of implementation of the Outline of the Plan for the GBA (the Outline), the GBA achieved remarkable results in various areas and demonstrated strong development momentum. Although the overall economy of the GBA shows a relatively healthy and high-quality development trend, there are obvious imbalances in regional economic development within the region, a situation that is widely known. Based on the effectiveness and outlook for 2035, how to promote HQED in the GBA is still an urgent issue that needs to be resolved. How can the HQED level in the GBA be measured and evaluated? Are there any unique spatial patterns and structural characteristics of the GBA in HQED? What are the key influencing factors driving HQED in the GBA? Clarifying these questions will help the central government formulate a more precise strategy for the GBA and provide valuable reference information for HQED in other regions.

Although research on the HQED of specific regions such as Beijing-Tianjin-Hebei, the Yangtze River Delta, and other urban agglomerations has been relatively abundant, empirical analyses related to the GBA are still lacking. Therefore, this study chose 11 cities in the GBA as the research object and constructed an evaluation index system for the level of HQED based on the five dimensions of “innovative development, green development, coordinated development, open development, and shared development”. To address the lack of existing studies on the high-quality development of the GBA in terms of spatial and temporal differentiation, we use the entropy weight method and standard deviation ellipse to explore the temporal evolution and spatial pattern development trend of the GBA. Furthermore, by analyzing the spatio-temporal evolution of HQED in the GBA in detail, we reveal its spatio-temporal difference characteristics as well as the current status of spatial agglomeration development and provide useful references and insights for other regions. By combining multidimensional methods such as the geographically weighted regression (GWR) method and geodetectors, we successfully overcome the shortcomings of traditional regression models in terms of quantitative attribution via spatial statistical methods. This enabled us to analyze the pathways of action of HQED in the GBA in greater depth, as well as the types of interactions among the developmental factors within it.

The subsequent sections of this paper include the following: the second part is a review of the relevant literature; the third part describes the research methodology, variables, and data collection; the fourth part analyses the analysis of the characteristics of spatio-temporal divergence and explores the driving mechanisms; and the fifth part sums the findings of the study and proposes policy recommendations.

2. Literature review

Clarifying the connotations of HQED is the primary issue in the study of HQED in urban areas. The connotation of the quality of urban development can be traced back as far as the idea of sustainable development in the 1992 United Nations Conference on Environment and Development’s adoption of Agenda 21 [4], and cities with HQED are characterized by social equity, environmental friendliness, and appropriate construction density [5]. Later, in the Istanbul Declaration issued by UN-HABITAT in 2011, it was noted that cities should be places that enable human beings to live with dignity, health, safety, happiness, and hope, providing a relatively clear definition and requirements for HQED [6,7]. High-quality development is the result of the optimization of economic structure, the transformation of old and new kinetic energy, the synergistic development of economy and society, and the significant improvement of people’s living standards after the growth of economic aggregate and scale to a certain stage [8,9]. In his research, Jin (2018) found that the productive forces and relations of production are constantly changing, and resource- and investment-driven development is gradually shifting to knowledge- and innovation-based sustainable development in the process of promoting high-quality development [5]. At the same time, it pays more attention to both domestic and international markets, continuously cultivates new advantages in international competition, and transforms the structure of supply and demand, the mode of governance, and ecological thinking by smoothly expanding the factor markets for land, capital, data, labor, and data and realizing the transformation of the national economy toward low input, high efficiency, high welfare, and sustainability [10]. Related studies by Yang et al. (2018) and Shen et al. (2020) noted that the high-quality development of an urban agglomeration economy currently faces notable challenges such as serious industrial isomorphism, inefficient resource allocation, a large gap between large, medium and small cities, and a synergistic governance mechanism [11,12]. The reasons for this are, on the one hand, the existence of administrative barriers and nonsharing of infrastructure among cities, resulting in policy communication as well as the sharing of public services seriously affecting the coordinated development of urban agglomerations; on the other hand, urban agglomerations are still in a stage of rapid development, and there are still large gaps in the industrial division of labor, resource allocation, and the smooth flow of factors compared with world-class urban agglomerations [5].

There is no uniform way to measure the level of HQED due to the large gap in the level of economic development between different countries and regions. Several international organizations have proposed representative systems of indicators for evaluation, such as the Human Development Index (HDI) proposed by the United Nations, which consists of life expectancy, level of education, and gross

national product per capita. The World Bank has constructed a system of world development indicators in its annual World Development Report, which covers all aspects of economic and social development, including economic growth, population, science and technology, education, the environment, climate change, etc. In addition, scholars have established a system of evaluation indicators based on the basic connotation of HQED, such as measuring the quality of economic development (QED) based on a single indicator, such as GDP [13,14]. Based on total factor productivity (TFP) and labor productivity, Chen et al. (2018), Hsieh et al. (2009), and Cubas et al. (2016) constructed multidimensional and multiindicator comprehensive evaluation bodies [15–17]. With the increasing role of green innovation in driving high-quality economic development, green total factor productivity is gradually becoming one of the indicators for measuring the quality of economic growth [18]. This indicator can comprehensively reflect the characteristics of urban development, the level of innovation, and the degree of sustainable development [19]. Since methods such as the DEA or SBM model of unexpected output used for total factor productivity cannot reflect the results of resource allocation and coordinated development and can underestimate capital accumulation and thus lead to inaccurate measurement of the high quality of the economy [20], a large number of studies have begun to measure high quality through the construction of a high-quality development evaluation index system. For example, the economic growth quality evaluation system includes three dimensions: “economic development, social life, and the ecological environment” [21,22]. Ou et al. (2020), based on new economic momentum, constructed the economic high-quality growth index from the four dimensions of technological innovation, structural reform, transformation, and upgrading and development benefits [23]. Alternatively, there is also a performance assessment system constructed from the perspective of industry, academia, and research [24]. The new development concept put forward by General Secretary Xi Jinping provides a unified index system for the evaluation of China’s high-quality economic development, enriching the connotation of high-quality economic development in five dimensions: economic power transformation, economic structure optimization, economic system stability, economic green development, and economic welfare sharing [16,25].

New economic geography points out that high-quality economic development is the result of the interaction of four major factors, namely, economies of scale, transportation costs, factor flows, and externalities, and is susceptible to the influence of various factors. From the perspective of economic growth theory, investment, consumption, and net exports are the “three carriages” that drive economic growth, while technological progress has a multiplier effect on economic growth [26,27]. Specifically, in the fields of economic and social development, industrial structure and technological upgrading are the core factors affecting high-quality economic development [28]. Zhang et al. (2021), Pan and Luo (2020), and Yang et al. (2021) argued that financial agglomeration and foreign direct investment affect the level of high-quality economic development by influencing the level of investment and enterprise profitability, respectively [29–31]. As one of the initiatives to boost national confidence and restore production and consumption in times of economic recession, infrastructure construction can indirectly affect HQED by promoting the upgrading of industrial structure and the effective allocation of resources. With the rapid development of digital technology, digital infrastructure determines the level of development of a country or region’s digital economy, and how to integrate the digital economy and traditional industries and optimize the industrial internet across borders has become an issue of concern for improving the quality of economic development [29, 31–33]. Wu et al. (2020) considered the incentives of environmental regulatory policies promoting regional coordinated development [34]. In addition, the level of education for human capital creation [35] and the degree of green investment in the context of carbon neutrality and peaking carbon [36] all play important roles in mitigating the “siphon effect”, improving the quality of human capital development, and promoting coordinated regional development [31,36].

Within the domain of research on HQED, a relatively systematic theoretical system has been constructed, and its connotation is very rich. However, the new development philosophy puts forward new requirements for HQED, there is still a deficiency in systematic empirical research, and its research framework and methodology show obvious deficiencies. First, the definition and measurement methods of HQED have not yet gained concurrence in the academic community, leading to significant differences in the research results of different scholars. In particular, we can see that scholars from different countries have different insights into the construction of an assessment system for high-quality economic development. Second, while studies on specific regions such as Beijing-Tianjin-Hebei, the Yangtze River Delta, and other urban groupings are more abundant, the GBA has a unique advantage that is incomparable to other regions in the HQED of China, and empirical analyses pertaining to HQED of the GBA have been insufficient. Third, the existing studies have focused more on qualitative exploration and traditional regression models, while the spatial correlation of HQED and its driving factors have not been thoroughly studied.

3. Study design

3.1. Indicators and methods for measuring HQED

HQED is the effective combination of economic development speed and QED. Based on the rich connotation of HQED, researchers have undertaken comprehensive studies on the construction of its evaluation indicator system and have implemented the five new development philosophies of “innovation-driven, coordinated, green, oriented toward global progress, and beneficial to all” proposed by General Secretary Xi Jinping to construct an evaluation index system for HQED. On the basis of existing research, based on the principles of scientificity, systematicity, practicality, and dynamism, and taking into account the availability and practical representativeness of the data, this paper constructs a high-quality development evaluation system from the following four levels to construct an evaluation index system for high-quality economic development:

First, at the level of development goals, “shared development” is the final goal of HQED. It is firmly embedded in the essence within the context of socialism with Chinese characteristics in the modern era, placing the people at the center of development and calling for the fruits of development to be shared with the people, fully demonstrating the idea of people-centered development.

The second is the development momentum layer, which contains two key subsystems, namely, “innovative development” and “open development”. On the one hand, “innovation and development” is at the center of the system structure and is a powerful driving force for HQED. As the first productive force of economic development, innovation has a multiplier effect on various factors of production, such as labor and capital, directly affecting the achievement of HQED goals and is the “golden key” for overcoming the bottleneck of HQED. On the other hand, “open development” is an important method to achieve HQED, focusing on addressing the issue of internal and external connections, with the target of enhancing the quality of opening up to the outside world, developing a higher-level open economy, and driving the formation of a high-level dual-cycle development model both domestically and internationally.

Third, at the dynamic level of development, “coordinated development” concentrates on resolving the issue of imbalanced development, such as the coordinated development of urban and rural regions, the economy and society, and material and spiritual civilization. By remedying shortcomings, strengthening the whole economy, and overcoming constraints, China enhances the balance, inclusiveness, and sustainability of development; promotes synergies and balanced development in all areas in all regions; and facilitates China’s economic and social stability and progress.

The fourth layer is the development constraint layer. Green development reflects the hard constraints in the course of HQED on the natural ecological environment and is a restrictive requirement for economic development and social activities. Green development requires adherence to a new path of industrialization with a strong emphasis on technology, robust economic efficiency, minimal resource consumption, low environmental impact, and the full utilization of human resources.

Based on the unique institutional advantages of the GBA, we selected the number of hospital beds, the pupil-teacher ratio in compulsory education, public library holdings and urban public transport vehicles as measures of “shared development”. We referred to the methodology of Yan et al. (2021) and selected R&D investment, number of R&D personnel, patent grants, number of publications and number of tertiary institutions to measure “innovation development” [37]. Referring to the methodology of Ma et al. (2019), total foreign trade and total actual utilization of foreign capital are chosen to measure “open development” [38]. We referred to the methodology of Zheng and He (2022) and selected GDP per capita, tertiary value added, the urbanization rate, the urban registered unemployment rate, the mortality rate of production safety accidents per 100 million yuan of GDP and the consumer price index increase to measure “coordinated development” [39]. We referred to the methods of Yan et al. (2021) and Ma et al. (2019) and selected the energy consumption, average annual fine particulate matter (PM_{2.5}) concentration, and financial environmental protection expenditure to measure “green development” [37,38].

3.2. Methodology for evaluating HQED

3.2.1. Entropy weight method

Referring to Yang et al. (2024) [40], Yang et al. (2023) [41], and Li et al. (2024) [42], taking into account the characteristics of the sample data and the objectivity of the weights of the indicators, this paper establishes an evaluation model of HQED through the entropy weight method, with n indicators and m cities, and the operation process of its calculation is as follows:

- (1) Constructing an evaluation matrix X_{ij} for HQED:

$$X_{ij} = \begin{Bmatrix} X_{11} & \cdots & X_{1n} \\ \vdots & \ddots & \vdots \\ X_{m1} & \cdots & X_{mn} \end{Bmatrix}, (i = 1, 2, \dots, m, j = 1, 2, \dots, n)$$

- (2) Normalizing the matrix of HQED X_{ij} yields the normalized matrix Y_{ij} :

$$\text{Positive indicators: } Y_{ij} = \frac{X_{ij} - \min(X_{ij})}{\max(X_{ij}) - \min(X_{ij})}, (i = 1, 2, \dots, m, j = 1, 2, \dots, n).$$

$$\text{Negative indicators: } Y_{ij} = \frac{\max(X_{ij}) - X_{ij}}{\max(X_{ij}) - \min(X_{ij})}, (i = 1, 2, \dots, m, j = 1, 2, \dots, n).$$

- (3) Calculate the information entropy e_j for each indicator:

$$E_j = -\frac{1}{\ln m} \sum_{i=1}^m P_{ij} \ln P_{ij}, P_{ij} = \frac{Y_{ij}}{\sum_{i=1}^m Y_{ij}}$$

- (4) Calculate the information redundancy d_j :

$$D_j = 1 - e_j, j = 1, 2, \dots, n$$

- (5) Calculate the weights w_j of the indicators:

$$W_j = \frac{D_j}{\sum_{j=1}^n D_j}, j = 1, 2, \dots, n$$

(6) Calculation of a composite score U for HQED:

$$U = \sum_{j=1}^n W_j Y_{ij}$$

This paper uses the entropy weight approach for determining the weight coefficients of each indicator. First, construct an evaluation matrix of HQED that contains m city samples on n evaluation indexes. Second, the evaluation matrix is standardized, and different standardization methods are adopted for positive and negative indicators. Once again, the information entropy for each indicator is computed to measure the extent of dispersion for that specific metric to reveal the contribution of each indicator to the evaluation results. Based on these information entropy values, the information redundancy is further calculated to reveal the proportion of effective information in each indicator. On the basis of the information redundancy obtained, the weight coefficients of each indicator are calculated to ensure that these weights are more objective and reasonable and can accurately reflect the influence of each indicator on the overall evaluation results (as shown in Table 1). Finally, based on the calculated weighting coefficients, a comprehensive score for HQED is derived. This score comprehensively reflects the performance of each city at the HQED level and is usually between 0 and 1. A higher score represents a higher level of HQED in the study population. Conversely, it indicates a lower level of development.

3.2.2. Gini coefficient

The Gini coefficient, developed on the basis of the Lorenz curve, has become one of the most widely used indicators in the academic world for judging the balanced level of regional development. To further judge the difference and concentration of the level of coupled and coordinated development of regional innovation ecosystems, the specific expression is as follows:

$$G = \frac{S_A}{S_{A+B}}, 0 \leq G \leq 1$$

where S_A is the area between the actual score index curve and the absolute average curve and S_{A+B} is the area between the absolute average curve and the absolute nonaverage curve. The smaller the Gini coefficient is the greater the level of regional equalization. When the Gini coefficient within a region is less than 0.2, regional development is highly equalized; when the Gini coefficient is greater than 0.5, the regional development gap is more disparate.

3.2.3. Trend surface analysis

Trend surface analysis applies the principle of regression analysis to approximate the actual surface, and the resulting trend surface can intuitively reflect the spatial distribution patterns and trends of the observed quantities. In this study, the trend surface was used to explore the overall spatial variation in HQED in the GBA. Suppose $Z_i(x_i, y_i)$ is the talent attraction value of city i , and (x_i, y_i) is the plane space coordinate, defined by the trend surface:

Table 1
System of indicators for evaluating HQED.

System Level	Weights	Normative Layer	Indicator Layer	Weights	Indicator properties
Innovative development	16.94 %	Innovative inputs	R&D investment	0.81 %	+
			Number of R&D personnel	3.66 %	+
		Innovation outputs	Patent grants	3.41 %	+
			Number of publications	4.65 %	+
			Number of tertiary institutions	4.41 %	+
Co-ordinated development	31.52 %	economic operation	GDP per capita	4.37 %	+
			Tertiary value added	4.46 %	+
			urbanization rate	3.65 %	+
		Urban and rural structures safe and stable	Urban registered unemployment rate	6.13 %	–
			Mortality rate of production safety accidents per 100 million yuan of GDP	3.64 %	–
Green development	15.31 %	consumer structure	Consumer price index increase	9.27 %	–
			Energy consumption	5.90 %	–
			Average annual fine particulate matter (PM2.5) concentration	5.96 %	–
		Energy reduction	Financial environmental protection expenditure	3.45 %	+
Open development	12.32 %	foreign trade	Total foreign trade	7.77 %	+
		utilize foreign capital	Total actual utilization of foreign capital	4.55 %	+
Shared development	23.91 %	medical level	Number of hospital beds	5.36 %	+
		cultural education	Pupil-teacher ratio in compulsory education	1.32 %	+
			Public library holdings	6.91 %	+
		infrastructure	Urban public transport vehicles	10.32 %	+

$$Z_i(x_i, y_i) = T_i(x_i, y_i) + \varepsilon_i$$

where $T_i(x_i, y_i) = \beta_0 + \beta_1 x + \beta_2 y + \beta_3 x^2 + \beta_4 y^2 + \beta_5 xy$ denotes the trend value over a wide range and ε_i is the autocorrelation random error, which denotes the deviation between the true value of talent attractiveness and the trend value for the i th city.

3.2.4. Standard deviation elliptic analysis

Standard deviation elliptic analysis (SDE) is a spatial statistical method that can present the overall characteristics of the distribution of economic geographic elements from multiple perspectives; i.e., it can describe the process of spatio-temporal evolution, which is usually expressed by using centrality, orientation, and shape characteristics. The spatial distribution ellipse of the elements consists of two spaces, the inner ellipse and the outer ellipse, and the main part is the part distributed inside the ellipse. An increase in the area of the ellipse, that is, when the spatial distribution ellipse shows an expanding trend, indicates that the growth rate of economic factors in the inner region of the ellipse is slower than that in the outer region of the ellipse; in contrast, a decrease in the area of the ellipse, that is, when the spatial distribution ellipse shows a contracting trend, indicates that the economic factors in the inner region of the ellipse are growing faster than those in the outer region of the ellipse. In this study, based on standard deviation elliptic analysis, the coordinates of the center point of the ellipse, the lengths of the long and short semiaxes, and their azimuths are analyzed to investigate the spatio-temporal evolution of the HQED pattern in the GBA. The specific formulas are as follows:

$$X = \sum_{i=1}^n D_i x_i / \sum_{i=1}^n D_i, Y = \sum_{i=1}^n D_i y_i / \sum_{i=1}^n D_i$$

$$\tan \theta = \frac{\left(\sum_{i=1}^n \tilde{x}_i^2 - \sum_{i=1}^n \tilde{y}_i^2 \right) + \sqrt{\left(\sum_{i=1}^n \tilde{x}_i^2 - \sum_{i=1}^n \tilde{y}_i^2 \right)^2 + 4 \left(\sum_{i=1}^n \tilde{x}_i \tilde{y}_i \right)^2}}{2 \sum_{i=1}^n \tilde{x}_i \tilde{y}_i} \delta_x = \sqrt{\sum_{i=1}^n (\tilde{x}_i \cos \theta - \tilde{y}_i \sin \theta)^2 / n}, \delta_y = \sqrt{\sum_{i=1}^n (\tilde{x}_i \sin \theta + \tilde{y}_i \cos \theta)^2 / n}$$

where X and Y are the coordinates of the center of HQED, x_i and y_i are the geographic coordinates of the cities, the ellipse azimuth angle θ is the angle formed by rotating clockwise to the long axis of the ellipse in the due north direction, $\tilde{x}_i$ and $\tilde{y}_i$ are the deviations of the geographic coordinates of each city from the center of gravity, and δ_x and δ_y are the standard deviations along the x-axis and y-axis, respectively.

3.2.5. Geodetector

Geodetector is a spatial analysis method for detecting spatial heterogeneity as well as exploring the driving forces behind it, which compensates for the limitations of traditional statistics where there are restrictions in the assumptions [41]. In this paper, we use the geodetector model to explore the influences and mechanisms of different factors and their interaction terms on the quality of development of the GBA. The formulas are as follows:

$$q = 1 - \frac{\sum_{h=1}^L N_h \delta_h^2}{N \delta_h^2} \quad (1)$$

$$\lambda = \frac{1}{\delta^2} \left[\sum_{h=1}^L \bar{Y}_h^2 - \frac{1}{N} \left(\sum_{h=1}^L \sqrt{N_h} \bar{Y}_h \right)^2 \right]$$

In the formula, N_h and δ_h are the unit value and variance of layer h , respectively, and the q value reflects the degree to which the factor explains the high-quality development of the economy. λ can test the significance of the q value.

3.2.6. Geographically weighted regression (GWR)

Geographically weighted regression (GWR) is a spatial analysis technique that is widely used to analyze spatial patterns in geography and related fields. GWR explores the spatial variation of a study object at a specific scale and the associated drivers by building a local regression equation at each point and can be used for the prediction of future results. It has higher accuracy because it takes into account the local effects of spatial objects. GWR extends the ordinary linear regression model by embedding the spatial location of the data into the regression equations with the following formulas:

$$y_i = \beta_0(\mu_i, v_i) x_{ik} + \varepsilon_i$$

where y_i is the value of the attribute of the explanatory variable at the i th sample point, x_{ik} is the value of the k th explanatory variable at point i , (μ_i, v_i) is the geographic coordinate at point i , and $\beta_k(\mu_i, v_i)$ is the regression coefficient of the first explanatory variable at the point. $\beta_0(\mu_i, v_i)$ is a constant term, and ε_i is a random error term. The adaptive quadratic kernel function and AICc criterion are often used as kernel functions and bandwidth selection criteria for GWR models.

3.3. Study objects and data sources

The GBA ($111^{\circ}21' - 114^{\circ}53'E$, $21^{\circ}28' - 24^{\circ}29'N$) is located on the southern coast of China and includes the two Special Administrative Regions of Hong Kong and Macao, as well as nine cities (Guangzhou, Shenzhen, Zhuhai, Foshan, Zhongshan, Jiangmen, Huizhou, Dongguan, and Zhaoqing) in Guangdong Province, with a total area of 56,098 square kilometers. As one of the engines of China's economy, the Guangdong-Hong Kong-Macao Greater Bay Area has been committed to realizing high-quality economic development. With a well-developed industrial system and superior clustering advantages, the region continues to promote innovation, green development, openness and cooperation and strives to achieve qualitative improvement in economic growth. By the end of 2022, the resident population of the GBA reached 86,290,400, accounting for 5.98 % of the country's total population. In 2023, the region's GDP totaled more than RMB 13.8 trillion, accounting for approximately 10.9 % of China's total GDP. HQED is a critical path for the GBA to help Hong Kong and Macao integrate into overall national development, to comprehensively expand opening up and cooperation with the outside world, to establish a globally influential international science and technology innovation hub and to become a new growth pole for the development of China and the world. This paper takes the 11 node cities in the GBA as the research object and spatial scope and uses the time period from 2011 to 2020 as the scope of the study. In addition, the data we obtained did not contain any missing values. In the process of data collection and collation, we found that some of the data had different statistical units. To ensure the consistency and comparability of the data, we converted all the data into the same statistical units. The data for the study are mainly obtained from the yearbooks of the cities in the PRD, the Guangdong Provincial Statistical Yearbook, the Guangdong Social Statistical Yearbook, the China Statistical Yearbook, the Hong Kong Statistical Yearbook, the Macao Statistical Yearbook, and the official data from the Hong Kong Census and Statistics Department and the Statistics and Census Service of Macao and the World Bank. The distances between cities are obtained from Baidu Maps.

4. Analysis of results

4.1. Time dynamics and equilibrium evolution of HQED in the GBA

4.1.1. Evolution of the time dynamics of HQED

To graphically and intuitively show the overall level, distributional evolution, ductility, and polarization trend of the HQED of the GBA, this paper plots the three-dimensional dynamic kernel densities of the HQED of the GBA, as shown in Fig. 2 (see Fig. 1). First, from the distribution position of the kernel density curve, the kernel density curve of HQED in the GBA exhibited an overall rightward shifting trend during the period of 2011–2020. This indicates that most of the cities in the GBA are on an upward trajectory in terms of their HQED level. Considering national policy support, technological innovation, industrial advancement, the attraction of foreign investment and trade cooperation, and infrastructure construction and collaborative urban development, it can be concluded that HQED in the GBA has achieved a stage-by-stage result. Second, in terms of the distribution pattern of the main peak, the kernel density curve from 2011 to 2018 shows that the height of the main peak decreased and its width increased. This means that the dispersion of the HQED of the city's economy exhibited an increasing trend throughout this period because the industrial bases of different regions are very different and there are large gaps in the development paths of various fields. The height of the main peak of the kernel density curve increased and decreased in width from 2019 to 2020, indicating that the absolute gap in the HQED level of these cities gradually narrowed. Finally, in terms of the number of peaks, the overall curve shows a double-peak phenomenon, but the peak value of the side

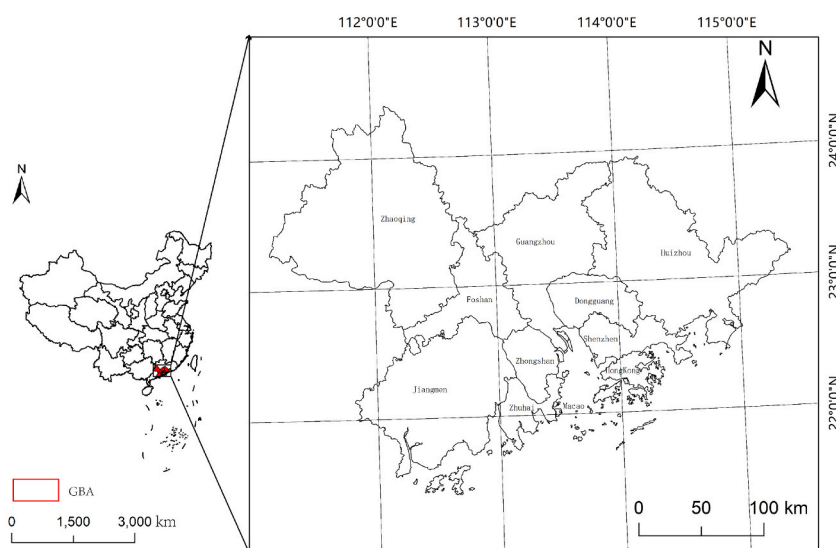


Fig. 1. Map of the GBA.

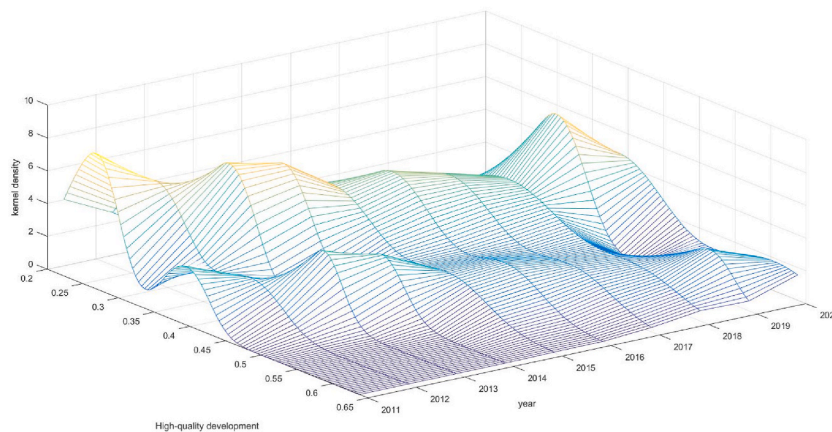


Fig. 2. Three-dimensional kernel density map of the QED scores in the GBA, 2011–2020.

peaks is lower, indicating that there is a weak polarization in the HQED level of the economy of the GBA.

4.1.2. Evolution of spatial equilibrium in the dimensions of HQED

The Gini coefficient can more intuitively show the degree of imbalance in HQED of the regional economy, so this paper measured the Gini coefficient of the comprehensive score of HQED from 2011 to 2020 and the Gini coefficients of the five dimensions. Table 2 shows that (1) the Gini coefficient of innovation development decreased from 0.598 in 2011 to 0.575 in 2020, which reflects a high degree of imbalance despite the decrease, indicating that there is a certain degree of imbalance in innovation resources, scientific and technological capabilities, and innovation inputs among different cities in the GBA. (2) The Gini coefficient for coordinated development displays a slightly fluctuating trend, declining from 0.099 in 2011 to 0.077 in 2020, indicating a relatively gradual reduction. This finding suggests that there is still some degree of disparity in coordinated development among regions. (3) The Gini coefficient of green development decreased from 0.191 in 2011 to 0.087 in 2020, indicating that the GBA has made remarkable progress in sustainable development and environmental protection. (4) The Gini coefficient of open development decreases from 0.704 in 2011 to 0.601 in 2020, reflecting a gradual reduction in the imbalance between different cities in the GBA in terms of outward openness and international collaboration. Relative to other normative strata, the Gini coefficient of open development is the highest, revealing the characteristics of differences in the conditions of international cooperation and outward-oriented economic models of different cities. (5) The Gini coefficient of shared development increases from 0.323 in 2011 to 0.326 in 2020, revealing relatively smaller imbalances in social equity and sharing. This shows that the GBA has achieved a certain degree of balance in social welfare, education, and employment. A comparison of the five guideline strata reveals that the differences in innovative, open and shared development are most prominent. Therefore, to achieve HQED in the GBA, it is vital to further strengthen the innovation-driven role of central cities, promote technology diffusion and knowledge spillover, coordinate innovation development on both the east and west sides of the Pearl River, and jointly create an international science, technology and innovation center of global influence through the integration of industry, academia and research. At the same time, it should continue to expand opening up, fulfilling the role of the pivot point of the GBA's domestic and international dual cycle and doing a good job of infrastructure development to realize the integrated development of the GBA's hard and soft connectivity.

Table 2
Gini coefficient results.

vintages	Composite score for HQED	Innovative development	Co-ordinated development	Green development	Open development	Shared development
2011	0.115	0.598	0.099	0.191	0.704	0.323
2012	0.120	0.584	0.099	0.159	0.689	0.322
2013	0.109	0.577	0.094	0.141	0.681	0.324
2014	0.110	0.569	0.094	0.133	0.686	0.333
2015	0.118	0.569	0.104	0.101	0.674	0.339
2016	0.104	0.567	0.084	0.107	0.621	0.340
2017	0.110	0.555	0.076	0.106	0.599	0.336
2018	0.118	0.550	0.079	0.102	0.591	0.345
2019	0.117	0.552	0.099	0.097	0.568	0.337
2020	0.122	0.575	0.077	0.087	0.601	0.326

4.2. Analysis of the spatial pattern of HQED in the GBA

4.2.1. Characteristics of spatial differentiation in HQED

To better study the time-space evolution characteristics of the HQED of the cities in the GBA, this paper takes 2011 as the base period and draws on the study of Ou et al. (2020), who classified the 11 cities into five types according to the characteristics of the HQED of the GBA, i.e., HQED cities (≥ 0.50), medium-quality development cities [0.40, 0.50), medium-quality developing cities [0.30, 0.40), medium-low quality developing cities [0.20, 0.30) and low-quality developing cities (0, 0.20), and visualized them in 2011, 2014, 2017 and 2020 as the comparative periods, as shown in Fig. 3 [23].

In terms of spatial patterns, the HQED of the GBA is characterized by “high in the middle and low in the periphery”. Cities with a high level of HQED are distributed on both sides of the Pearl River, with a spatial structure of “core-periphery”, forming a high-value zone with Guangzhou, Shenzhen, and Hong Kong at its core, while the HQED level of peripheral cities is relatively low. In addition, the number of cities with both HQED levels and HQED levels increased significantly between 2005 and 2019. From the only HQED city in 2010, the number reached six cities in 2020, with Guangzhou having the fastest rate of HQED, increasing from 0.395 to 0.638.

In terms of the temporal evolution of the spatial distribution, the HQED in the GBA was generally low in 2011, with the proportion of low-quality and medium-low-quality development cities accounting for 72.73 %, or a total of eight cities. The reason may be that in the early days, these cities overpursued economic growth and practiced a crude development model, with very little awareness of HQED. Insufficient attention has been given to improving the capacity for coordinated urban development under this economic model, leading to a lower level of integrated development in various regions, which ultimately inhibits the improvement of the level of HQED. In 2014, the level of HQED of the cities was significantly greater than that in 2011, with Guangzhou (0.472) and Shenzhen (0.458) leaping to become medium- and HQED cities, but in general, the level of HQED of the GBA is still to improve. In 2017, Guangzhou (0.540), Shenzhen (0.517), and Hong Kong (0.528) again significantly increased their level of HQED, jumping from medium-quality developing cities to HQED cities. The level of HQED in each city increased significantly, and the number of cities with medium-quality development and HQED types reached four, with Huizhou (0.403) also becoming a medium-quality development city. In 2020, Foshan (0.421) and Dongguan (0.409) joined the medium-quality development gradient, and the HQED levels of the 11 cities in the GBA increased.

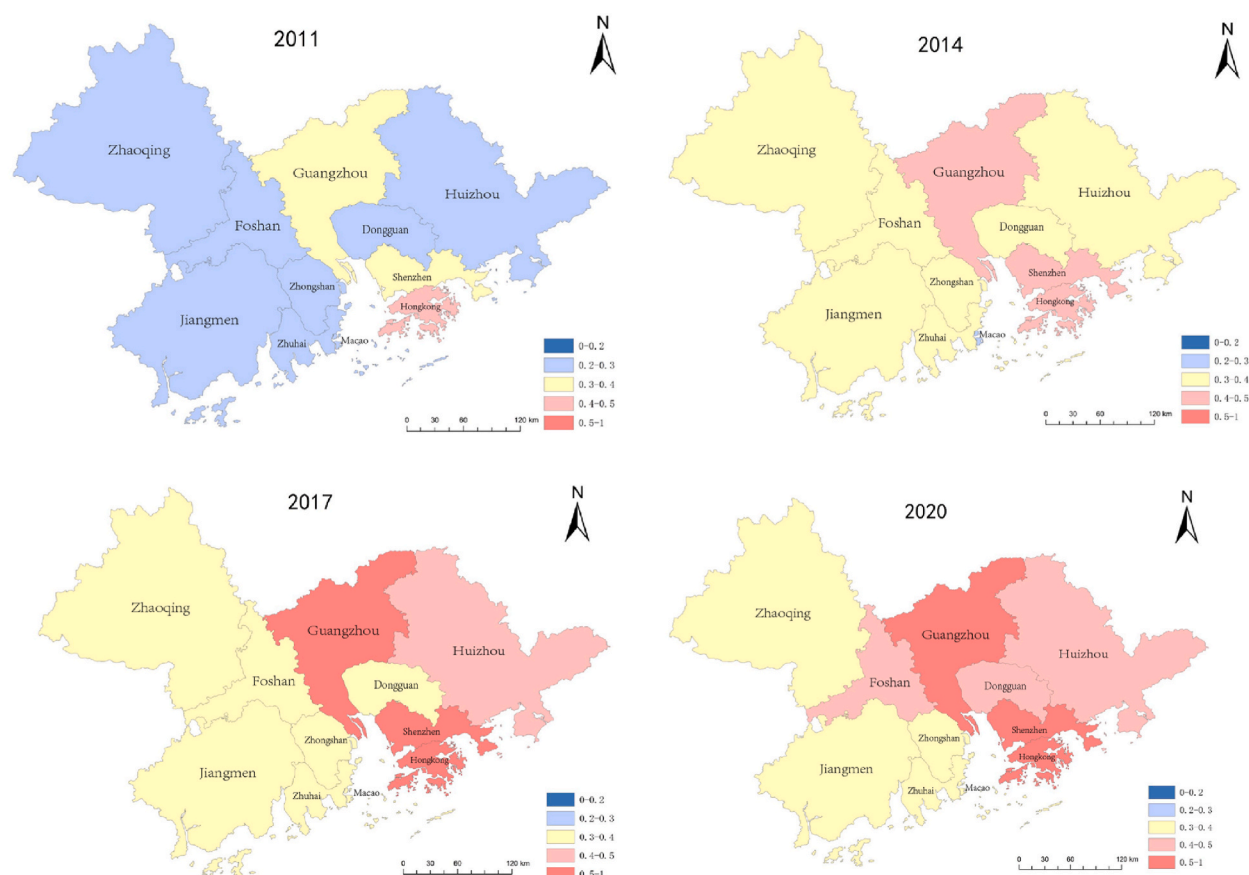


Fig. 3. Spatial distribution of the HQED level in the GBA.

4.2.2. Center of gravity and standard deviation ellipse evolution characteristics

The general features of the spatial distribution of economic factors may also vary at different temporal or spatial scales. With the aim of accurately determining the spatial distribution pattern of HQED in the GBA, this paper will focus on the process of spatial and temporal changes in the pattern of HQED and analyze the evolution pattern of the spatial distribution of HQED in the GBA. Using ArcGIS software, the SDE method was employed to gauge the level of HQED of the GBA in 2011, 2014, 2017, and 2020, and the results are presented in Table 3.

Table 3 shows that the spatial and temporal evolution of HQED in the GBA from 2011 to 2020 is characterized by the following features (see Table 4). First, in terms of the coordinates of the center of gravity, it shifted to the northwest between 2011 and 2014, highlighting HQED in the northwestern part of the GBA during that period. The center of gravity then shifted to the southeast, highlighting the HQED of the southeastern part of the region during this period, with special emphasis on the high-quality synergistic development of the Shenzhen-Hong Kong cluster, while the HQED of the western and northern parts of Guangdong was relatively slower. Second, in terms of the shape of the distribution, the azimuth angle and the long and short semiaxes of the standard deviation ellipse of the spatial distribution of HQED in the GBA changed only to a limited extent during the period from 2011 to 2020. The azimuth angle shows a slow upward trend from 100.979° in 2011 to 101.996° in 2020, indicating that the ellipse rotates clockwise. The long semiaxis is consistently larger than the short semiaxis, with the long semiaxis increasing and then decreasing, while the short semiaxis remains basically stable. This indicates that the spatial directionality of HQED in the GBA is more obvious. Third, in terms of spatial scope, the area covered by the standard deviation ellipse of the spatial distribution of HQED in the GBA experienced an increase and then a decrease from 2011 to 2020. From 2011 to 2017, the area covered increased from 18,460,000 square kilometers to 18,800,000 square kilometers, which is an expanding trend. However, by 2020, the area covered has decreased to 18.44 million square kilometers, and the coverage of the standard deviation ellipse has gradually shrunk to the southeast, indicating that the capacity for HQED in the internal areas of the GBA has increased at a significantly faster rate than that of the external areas and that the rate of increase has been more pronounced in the cities in the southeast direction in particular.

4.2.3. Spatial trend surface analysis

To present more clearly the overall spatial evolution trend and distribution law of HQED, this paper takes the comprehensive HQED scores of the four time nodes of 2011, 2014, 2017, and 2020 and makes use of the trend surface analysis tool in ArcGIS software to express the spatial visualization of HQED through smoothed mathematical surfaces, as shown in Fig. 4.

Overall, in the north–south direction, an inverted “U” curve is evident, with the central high, the sides low, and the sides higher in the North than in the South, and the slope and curvature increase daily, with the apex progressively shifting to the middle. Guangzhou, which scored high in terms of HQED, is located in the north of the center. As the level of HQED in Shenzhen and Hong Kong increases, north–south differences are narrowing, and the regional balance somewhat improves. The trend line is oriented diagonally in the east–west direction, indicating that the HQED level in the GBA from 2011 to 2020 exhibits a spatial distribution pattern of being high in the east and low in the west. The trend of upward development in the central region is becoming more pronounced, transitioning from a smooth curve to a progressively stronger parabola.

Specifically, the HQED level in 2011 was characterized by a high level in the east and a low level in the west and a high level in the south and a low level in the north. This indicates that there is spatial differentiation in the HQED of the GBA in the east–west and north–south directions; compared with 2011, the trend surfaces of the center and the east in 2020 will be nearly straight-line, and the trend surfaces in the north–south direction will become smooth. This indicates that HQED in the GBA has narrowed the gap between the north and south due to the development of the central region, but there is still spatial differentiation. Therefore, the curve always maintains the transmutation trend of bulging toward the center, both in the north–south and east–west directions. The HQED of the GBA has obvious spatial directionality. The HQED levels of the cities located in the periphery of Guangzhou, Shenzhen, and Hong Kong are increasingly gaining prominence, making them advantageous spaces for HQED in the GBA.

4.3. Detection of drivers of HQED

In order to deeply analyze the intrinsic mechanism and driving factors of HQED, this paper adopts the geodetector model. The model does not require strict conditional assumptions, can effectively overcome the problem of multivariate covariance, and analyses the interactions between intrinsic drivers and their interactions by measuring spatial heterogeneity [43]. The model combines GIS spatial overlay technology and set theory with clear physical meaning to accurately identify the relationships between multiple factors [44].

In terms of variable selection, agglomeration level, human capital, foreign investment, infrastructure construction, financial environmental protection expenditures, and fiscal science and technology investment were selected as the core driving factors derived

Table 3
Elliptic coefficients of the standard deviation of HQED in the GBA.

Year	Coordinates of the center of gravity	Long half shaft (km)	Short half shaft (km)	Azimuth (degrees)	Area (10,000 km ²)
2011	113.538°E , 22.768°N	0.458	0.321	100.979	1.846
2014	113.536°E , 22.785°N	0.466	0.320	101.382	1.874
2017	113.537°E , 22.782°N	0.466	0.321	101.857	1.880
2020	113.539°E , 22.781°N	0.459	0.320	101.996	1.844

Table 4
Detection of factors for HQED.

impact factor	Value of q-statistic	p-value
Level of agglomeration X_1	0.459	0.00
Level of human capital X_2	0.517	0.00
Level of foreign investment X_3	0.546	0.00
Transport infrastructure X_4	0.189	0.00
Financial environmental protection expenditure X_5	0.496	0.00
Financial investment in science and technology X_6	0.446	0.05

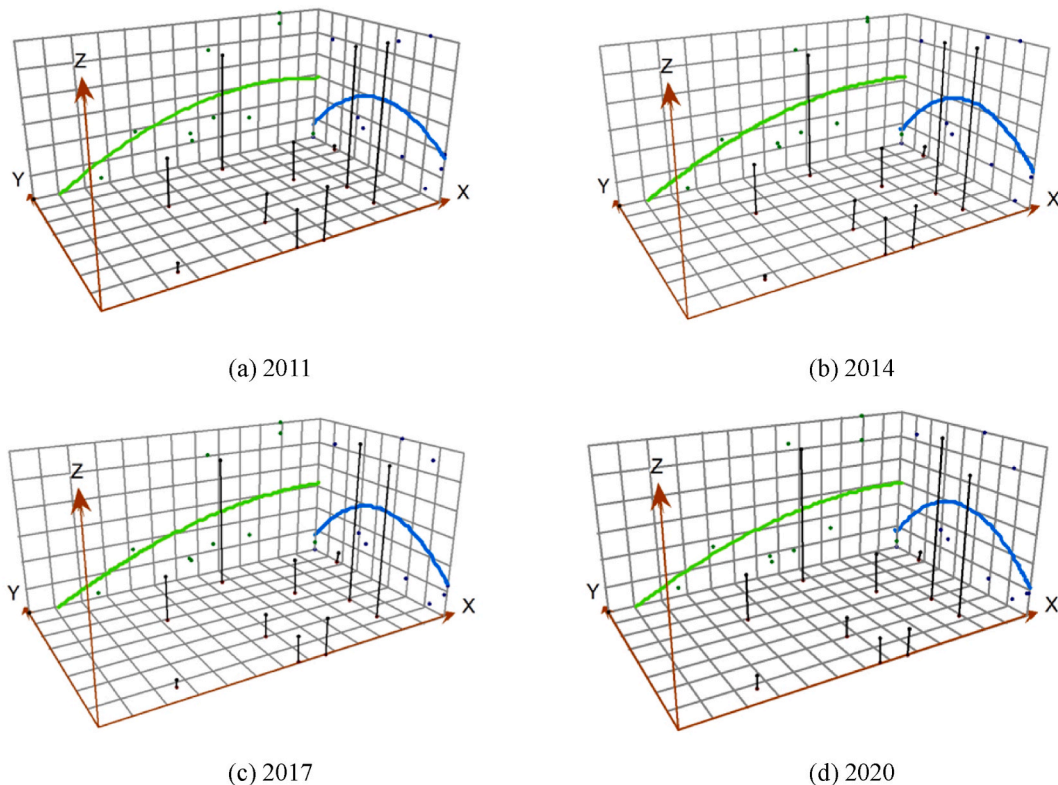


Fig. 4. Trend surface analysis of HQED in the GBA.

from an analysis of the influencing mechanism. Specifically, agglomeration level is measured by population density; human capital is represented by the enrollment figures of students in colleges and universities; foreign investment is quantified by the total utilized amount of foreign investments; the level of transport infrastructure is expressed by road density (the ratio of road mileage to the area of administrative divisions); and the level of financial environmental protection expenditures and fiscal and scientific and technological inputs are quantified by financial environmental protection expenditures and fiscal and scientific and technological allocations, respectively.

4.3.1. Detection of factors for HQED

The natural discontinuity method is a statistical method of classification according to the statistical distribution pattern of numerical values, which can appropriately group similar values to maximize the differences between groups. According to the statistical distribution law of each index, this paper employed the natural discontinuity method to classify the similar values of five variables from 2011 to 2020 into five levels, maximize the differences between the groups, realize the conversion of numerical data into typological data, import the classification results into the geodetector, and obtain the results of the factor detection, as shown in Table 3.

According to the regression results of the factors influencing HQED in each of the 11 cities in the GBA, all five variables significantly drive HQED in the GBA, and all of them are significant at the 1 percent and 5 percent levels. First, the level of agglomeration (q-value of 0.459) significantly drives HQED. Population agglomeration promotes technological progress, and population mobility facilitates the entry of low-income people into high-level cities, which is beneficial for reducing income disparities and synergistic development among cities. It ranks third in the explanatory power of HQED, passing the 1 % significance level, indicating that the higher the level of

agglomeration is, the more it can exert the effect of economies of scale and realize knowledge spillover to promote HQED in urban areas. Second, the improvement of human capital (q-value of 0.517) also significantly promotes HQED. The competition for high-level talent becomes a new driving force, and the improvement of human capital boosts the optimization of production efficiency and industrial structure. The level of foreign investment (q-value of 0.546) ranks first in terms of explanatory power, showing a significant contribution. At the same time, foreign investment injects capital and technology to promote resource utilization efficiency and innovation capacity. Again, transport infrastructure (q-value of 0.189) has less explanatory power but still passes the 1 % significance level, and its facilitating role cannot be ignored. Transportation facilities reduce the cost of production factors, promote resource allocation, and help economic growth and urbanization. Again, fiscal environmental protection expenditure (q-value of 0.254) is significantly beneficial to HQED, favoring ecological environment construction and attracting high-quality enterprises and talent. Finally, fiscal investment in science and technology (q-value of 0.446) can significantly improve the level of HQED. This indicates that the development of the GBA is gradually shifting to a knowledge-driven development mode, increasing investment in science and technology innovation and accelerating the construction of an international science and technology innovation center.

4.3.2. Interaction detection for HQED

The geo-detector can identify the actual interplay among factors, encompassing their intensity, direction, linear correlations, etc., and are not limited to predetermined multiplicative interactions, as determined by econometrics [43]. Furthermore, interaction detection is employed to assess how the interactions between various factors impact coupled coordinated development, and the results are detailed in Table 5. The findings indicate that the combined impact of any two influencing factors following interaction surpasses the individual effect of a single factor, indicating that the HQED of the GBA is the result of the joint action of multiple factors and that the HQED of the GBA has formed a benign composite mechanism for the coordinated development of the region. The HQED of the GBA has formed a benign compound regional coordinated development mechanism.

Comprehensive analyses show that the interaction values of the two indicators of foreign investment level and financial investment in science and technology with other factors are all greater than 0.70, which effectively highlights the superposition and multiplier effects of foreign investment and science and innovation support. This further shows that the GBA has the advantage of gathering global innovation resources, is the frontier of China's development of an open economy, and has a unique advantage as an international science and technology innovation center. Financial investment in science and technology is the material support of innovation activities, and adequate financial support helps to improve the efficiency of scientific research and provides a guarantee for the optimization and upgrading of other factors. After this interaction, fiscal investment in science and technology can increase R&D investment, optimize the efficiency of resource use, attract more high-level talent to the GBA, expand regional innovation demand to the global scale, establish a global vision, draw on international experience in science and technology innovation, and create an inclusive, harmonious and open innovation ecological environment. In particular, the maximum interaction value between the level of agglomeration and financial investment in science and technology reaches 0.809, showing that the interaction between these two factors has a significant impact on the spatial difference in HQED in the GBA. A business environment with sufficient scientific research funding and close communication with the international community promotes the optimization and integration of human capital, accelerates the cross-border flow of resource elements and technology diffusion, and promotes the accelerated transformation of R&D results rooted in basic R&D and pilot transformation, driving the improvement of the scientific and technological level of the GBA and even the whole country with innovation.

Compared with the results of the single-factor probes, the interaction of the factors in each dimension had a significantly greater role in driving the spatial variation in the level of HQED of the GBA than that of a single factor. In particular, the ability of transport infrastructure to explain the HQED of the GBA significantly increased after it interacted with other factors. This suggests that the continuous promotion of transport infrastructure construction in the GBA, the improvement of comprehensive transport corridors, the creation of integrated transport hubs, and efforts to build an interconnected and comprehensive three-dimensional transport network can support the role of multiple factors in promoting HQED. In particular, the interaction value of transport infrastructure and

Table 5
Detection of interactions for HQED.

Interact in pairs	Interaction value	Comparison value	Interactive results
$X_1 \cap X_2$	0.770	$> \max[q(X_1), q(X_2)]$	two-factor enhancement
$X_1 \cap X_3$	0.639	$> \max[q(X_1), q(X_3)]$	two-factor enhancement
$X_1 \cap X_4$	0.555	$> \max[q(X_1), q(X_4)]$	two-factor enhancement
$X_1 \cap X_5$	0.632	$> \max[q(X_1), q(X_5)]$	two-factor enhancement
$X_1 \cap X_6$	0.809	$> \max[q(X_2), q(X_3)]$	two-factor enhancement
$X_2 \cap X_3$	0.632	$> \max[q(X_2), q(X_4)]$	two-factor enhancement
$X_2 \cap X_4$	0.598	$> \max[q(X_2), q(X_5)]$	two-factor enhancement
$X_2 \cap X_5$	0.772	$> \max[q(X_3), q(X_4)]$	nonlinear enhancement
$X_2 \cap X_6$	0.799	$> \max[q(X_3), q(X_5)]$	two-factor enhancement
$X_3 \cap X_4$	0.703	$> \max[q(X_4), q(X_5)]$	two-factor enhancement
$X_3 \cap X_5$	0.713	$> \max[q(X_3), q(X_5)]$	two-factor enhancement
$X_3 \cap X_6$	0.769	$> \max[q(X_3), q(X_6)]$	two-factor enhancement
$X_4 \cap X_5$	0.651	$> \max[q(X_4), q(X_5)]$	two-factor enhancement
$X_4 \cap X_6$	0.653	$> \max[q(X_4), q(X_6)]$	nonlinear enhancement
$X_5 \cap X_6$	0.679	$> \max[q(X_5), q(X_6)]$	two-factor enhancement

financial, scientific and technological investment exceeds the single factor influence of the two drivers, showing nonlinear enhancement. This indicates that the construction of infrastructure needs to be intensified along with technological innovation, and a perfect infrastructure provides an efficient and convenient innovation environment, which leads to the diversification of cooperative choices of innovation subjects, promotes healthy competition, and further optimizes the path of HQED in the GBA.

4.3.3. Analysis of GWR influences on HQED

The GWR model is additionally introduced to conduct a deep analysis of the localized spatial variations in the strength and direction of each factor's influence on HQED. After testing, the corresponding R^2 under the OLS model estimation is 0.969, while the R^2 under the GWR model estimation is 0.995, which shows that the model fitting effect using GWR is better. Fig. 5 shows a visual representation of the results of the regression coefficients of the six explanatory variables. The regression results fluctuate between positive and negative, the overall spatial distribution is nonstationary, and the spatial effect of each factor on HQED is obvious, with large differences in the degree of variability and distribution characteristics. In terms of the degree of impact of the explanatory

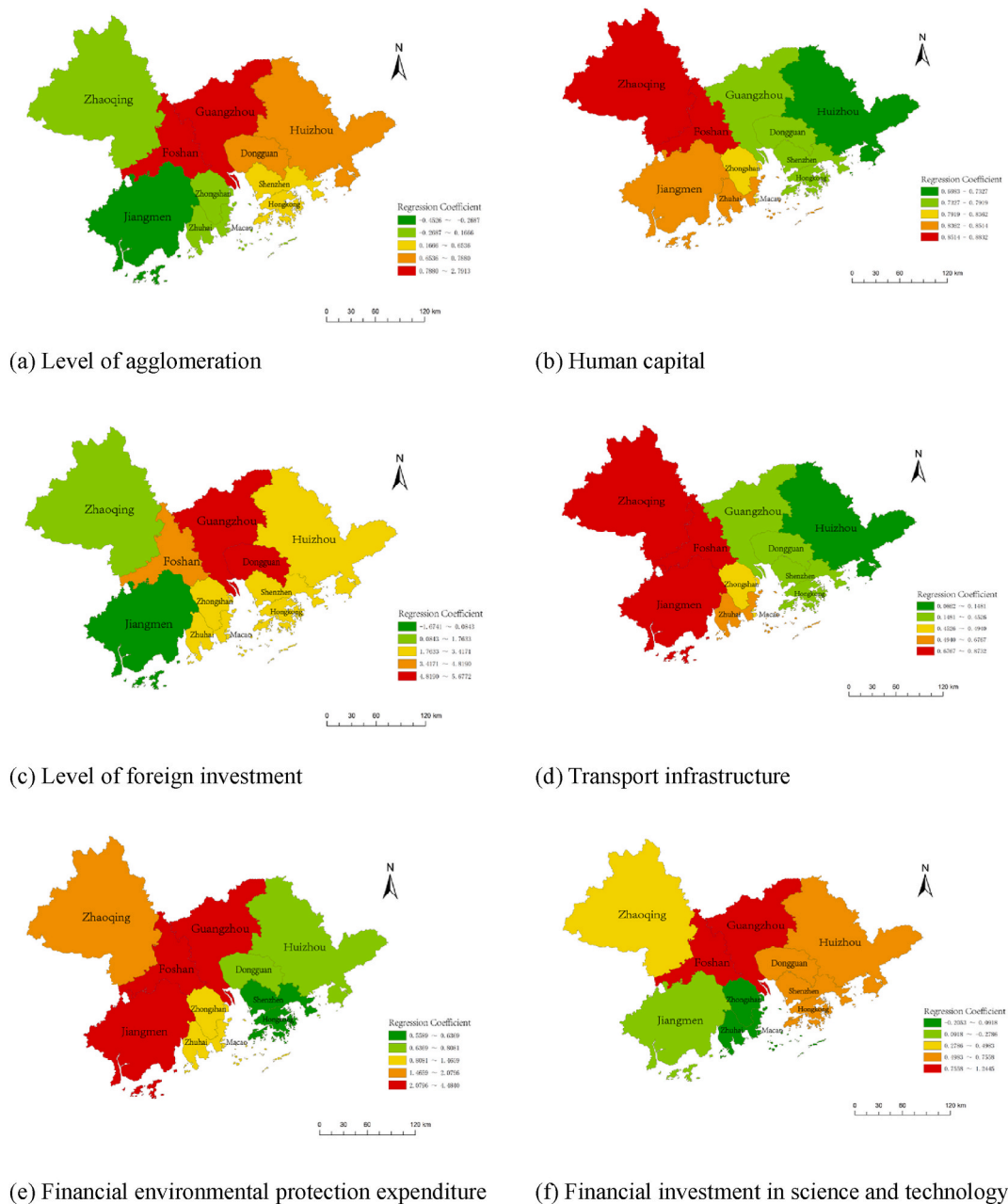


Fig. 5. Spatial distribution of the regression coefficients of the influential factors in the GWR model.

variables, in descending order, they are the level of foreign investment, fiscal environmental protection expenditure, agglomeration level, fiscal science and technology investment, human capital, and transport infrastructure.

Specifically, first, the agglomeration level has different effects on HQED ($-0.4526\sim2.7913$), and the spatial difference is more significant; the agglomeration level of Guangzhou and Foshan has the strongest driving effect on HQED, and the economic effect of the population agglomeration effect effectively promotes HQED in the GBA. Second, the regional influence of human capital has the smallest difference ($0.6983\sim0.8832$), demonstrating an overall pattern of “high in the west and low in the east. Moreover, the enhancement of human capital has an obvious positive effect on HQED in the GBA. Third, the spatial heterogeneity of the regression coefficient of the foreign investment level is more obvious ($-1.6741\sim5.6772$), illustrating the features of “elevated values in the central area with lower values in the surroundings and higher values in the northern portion with lower values in the southern portion”. The northern and central parts of the GBA have the strongest reliance on foreign investment, with Guangzhou and Dongguan being the most significant. Zhaoqing has a weak negative effect, which is more consistent with the previous state of Zhaoqing’s HQED, indicating that the driving effect of the level of foreign investment in Zhaoqing has difficulty effectively driving the city’s HQED. Fourth, the effects of transport infrastructure ($0.0862\sim8.8732$) and financial and environmental protection expenditure ($0.5589\sim4.4840$) on HQED have more pronounced positive feedback effects, and the overall region shows a “high west and low east, from the west to the east in decreasing order” distribution pattern. Fifth, the impact of financial investment in science and technology is more balanced ($-0.2053\sim1.2445$), with the overall presentation of “high in the northeast, low in the southwest” promotion effect. Among them, the region with the smallest regression coefficient coincides with the low-value agglomeration area of HQED. This suggests that financial investment in science and technology does not necessarily promote HQED in disadvantaged cities but may lead to the siphoning off of human and technological resources, which further widens the regional differences in HQED in the GBA.

5. Discussion

5.1. Innovative points

This study delves into several key aspects of HQED in the GBA and aims to broaden the research horizons for quality economic development in geography:

- (1) Constructing a comprehensive evaluation index system: This paper provides a new assessment framework for the HQED of the GBA based on five new development concepts (innovative development, coordinated development, green development, open development and shared development). This system not only overcomes the limitations of previous studies that favored qualitative exploration and traditional regression models but also fully considers the spatial correlation of high-quality development and its driving factors and comprehensively deepens the regional differences and qualitative characteristics to improve the accuracy and comparability of the study.
- (2) In-depth exploration of spatio-temporal evolution characteristics: This study fills the gap in existing studies in spatio-temporal differentiation. By analyzing the spatio-temporal evolution of economic development in detail, this study reveals the spatio-temporal differentiation characteristics of HQED in the GBA and the current status of spatial agglomeration development. This study provides useful references and insights for other regions.
- (3) Innovative methodology for analyzing the mechanism of regional high-quality development: This paper adopts an innovative methodology to make a breakthrough in the analysis of the mechanism of regional high-quality development. The shortcomings of traditional regression models are overcome through the use of quantitative spatial statistical methods. By combining multidimensional methods such as the GWR method and geo-detectors, the mechanism of the driving factors such as agglomeration level, human capital, foreign investment level, infrastructure construction, financial and environmental protection expenditures, and financial and scientific and technological inputs are explored in depth, and the geographic heterogeneity of the driving factors of high-quality development as well as synergistic or antagonistic effects among the factors are revealed, which provides a new analytical tool and idea for this field.

The integration of the above research strengths not only profoundly expands the theoretical foundation of HQED in the GBA but also provides rich insights for future related research.

5.2. Theoretical contributions

This paper makes two theoretical contributions:

The GBA is used as a research object to explore the spatial structure and driving factors of its high-quality development. As scholars from different countries have different insights into the assessment system of HQED, this helps the GBA create a model area for China’s high-quality development and provides insights into the high-quality development of other regions in China and even other regions around the world.

The enrichment of the theoretical connotation of HQED. HQED consists of two dimensions, “quantity” and “quality”, where “quality” usually refers to the structure and efficiency of economic development, and “quantity” usually refers to the scale and speed of economic development. Quality enhancement provides a sustained impetus for quantitative growth; quantitative growth provides an important foundation for quality enhancement, and the two are mutually reinforcing. In this paper, we continue to characterize the two dimensions of “quality” and “quantity” and construct an evaluation index system from the five aspects of “innovative

development, green development, coordinated development, open development, and shared development” to enrich the connotation of “quality” and “quantity” of HQED. Among them, innovation development is a strong driving force in the high-quality development of the economy, which has a multiplier effect on various factors, such as labor and capital, and provides a “gas pedal” for the realization of “quantity” while ensuring “quality”. Green development is an inevitable requirement for the high-quality development of the economy, which means that high-quality development should be anchored in the goals of high value-added, energy conservation and emission reduction, efficient use of resources, and green innovation, which is the ultimate goal of “quality” and the constraint of “quantity”. Coordinated development is the “balancer” of HQED, focusing on narrowing the gap between urban and rural, regional, economic and social development and ensuring the realization of “quantitative” equity and “qualitative” sharing. Open development is an important method of HQED and is the key step in opening up the domestic and international double cycle, not only from “quantity” at the standard international level but also from “quality” to catch up and surpass. Shared development is the core essence of HQED, which emphasizes that the fruits of development are to be shared by all people, and even more so, it calls for all people to work together to achieve individual quality while pursuing quantity. Therefore, this paper theoretically explores the theoretical connotations of realizing the “quality” and “quantity” of HQED based on the five dimensions of innovation, greenness, coordination, openness, and sharing and improves and supplements the existing macrofeatures of HQED.

The theory related to regional space was enriched, and the evolutionary characteristics of regional space were demonstrated through the use of cross-disciplinary methods. In this paper, multidisciplinary tools from human geography, physics, spatial economics, and urban and rural planning, such as spatial Gini coefficients, standard deviation ellipses, spatial trend surfaces, geodetectors, and GWRs, are utilized not only to present spatial equilibrium characteristics and spatial morphological changes in the HQED level of the GBA but also to detect the factors influencing the HQED level of the GBA and the interactions among the factors. These studies on spatial characteristics and spatial impacts are difficult to realize only by relying on the empirical tools of economics, and the embedding of relevant cross-disciplinary tools and methods better reveals the spatial characteristics and spatial evolution laws of regional development. This is also advocated by the new economic geography, where spatial factors play an important role in regional high-quality development. This paper describes both a spatial expansion of traditional regional economics and a practical application of new economic geography.

5.3. Policy recommendations

Drawing from the relevant discoveries in this paper, subsequent suggestions are proposed.

In promoting HQED in the GBA, the concept of win–win regional cooperation should be emphasized, transcending the traditional concept of development centered on local interests. With the aim of making full use of the significant advantages of cities such as Guangzhou, Foshan and Shenzhen, the overall radiation effect on other cities should be strengthened, especially to drive the development of neighboring cities such as Zhaoqing, Huizhou and Jiangmen, to realize the free flow of human capital, technological advantages and the efficient allocation of material capital within the city cluster. This will help the 11 cities in the GBA become a community of interest, smooth the free flow of production factor resources within the GBA, narrow the development gap between the peripheral cities and the central cities, and break through the bottleneck of HQED in the GBA.

Innovation is a core driver of HQED. Financial investment in science and technology and the building of an innovation ecosystem need to be strengthened, and the technological innovation power of enterprises and research institutes should be enhanced to achieve independent innovation in key areas. The development of urban agglomerations requires autonomous mastery of high and new technologies, particularly to capitalize on the innovative strengths of Hong Kong and Macao. To promote innovation, institutional reforms are needed to create a favorable environment for the effective implementation of scientific research projects and the establishment of a mechanism for the training of innovative talent. The GBA should strengthen the introduction of high-level talent, establish a mechanism for exchanging talent among cities, optimize the allocation of talent, formulate differentiated introduction policies, attract top global talent and jointly promote high-quality development.

The HQED of the GBA relies on the dual promotion of transport infrastructure and green development strategies. First, improving transport infrastructure is a key factor in promoting the effective allocation of resources, supporting industrial layout and attracting talent. The GBA should make full use of its geographical advantages, strengthen infrastructure planning for roads, railroads and airports, build a modern transportation network, achieve regional connectivity, develop ports and enhance its economic radiance. At the same time, financial expenditures on environmental protection should be increased to achieve sustainable green development. It is necessary to integrate resource environmental protection planning, strengthen the protection of natural resources, strengthen legal rewards and punishment mechanisms, enhance environmental awareness and reduce environmental damage. Actively develop the new energy industry, increase the share of the industry, promote the expansion of the scale of the new energy automobile industry, and promote the green development of the economy.

The GBA plays an important role in the Belt and Road Initiative. In particular, Hong Kong and Macao, as the key windows connecting the Mainland and the countries along the Belt and Road, play a significant role in the exchanges and cooperation between PRD cities and international enterprises. First, the “Belt and Road” should be used as an opportunity to take advantage of the three pilot FTZs in Qianhai Shekou of Shenzhen, Nansha of Guangzhou and Hengqin of Zhuhai to innovate the mode of cooperation, connect the international financial and shipping center of Hong Kong and the commercial and trade service platform of Macao, and give full play to the positive role of the GBA as the gateway of opening up to the outside world under the “Belt and Road”. The GBA should also tailor its policies to local conditions, formulate targeted incentives to attract the participation of foreign enterprises, and actively interface with international rules and standards. The government should make full use of the effects of the technology and human capital brought about by foreign investment, encourage independent innovation and basic research and development, and enhance the strength of the

GBA under high-level global competition to help realize HQED goals.

6. Conclusion

Studying the spatio-temporal differentiation of HQED in the GBA and its driving mechanism and clarifying the differences in HQED in the GBA can provide a scientific basis for promoting the coordinated development of the GBA. This paper tries to clarify the intrinsic influence mechanism of HQED at the theoretical level by constructing an indicator system, measuring the spatio-temporal evolution of the HQED capacity of 11 cities in the GBA from 2011 to 2020 using the entropy weight method, and detecting the driving factors by applying the geodetector model. The main conclusions are as follows:

There are three major characteristics of HQED in the GBA: a continuous upward trend over time, an obvious “Matthew effect”, a difference in the HQED level in the GBA characterized by a “low level of competence and a small gap in the overall situation”, and spatial evolution that is characterized by “high north and low south, high east and low west”. Spatially, the center of gravity of the high economic development ellipse of the GBA is always located in Guangzhou, with obvious spatial directionality, and the degree of spatial agglomeration is gradually increasing, showing a pattern of distribution in a southeast–northwest direction. and the coverage of the standard deviation ellipse gradually narrows toward the southeast. The analysis of the driving factors revealed that the level of agglomeration, human capital, the level of foreign investment, infrastructure development, financial and environmental protection expenditures, and financial, scientific and technological inputs positively affected HQED in the GBA, and the interaction effect of these two factors was prominent; in particular, infrastructure development and financial inputs, and scientific and technological inputs mutually reinforced HQED. There are geographical differences in the feedback effects of different cities, with human capital and infrastructure contributing positively, while the level of agglomeration and the level of foreign investment have different effects.

This paper establishes a system of evaluation indicators for HQED and conducts a penetrating investigation into its spatial and temporal evolution, along with the influencing factors, which comprehensively and objectively recognizes the current situation of HQED in the GBA and the problems it faces and provides references to promote the enhancement of the level of HQED in the GBA. However, limited by the research conditions and vision, this study still has many shortcomings, and more in-depth research is needed in the future:

In terms of research limitations, although the evaluation index system for HQED in the GBA established in this paper is relatively perfect, it is still difficult to fully reflect the reality of the utility due to factors such as the different statistical calibres of the three places and the difficulty in obtaining data. Future research will use Python tools to obtain open source city information data to supplement the indicator system. In addition, the scope of the study in this paper is limited to a single city cluster in the GBA, without expanding the focus to regions such as Beijing-Tianjin-Hebei and the Yangtze River Delta or analyzing it on a national or global scale. A more comprehensive study should cover global, national and local scales and analyze different units to obtain more comprehensive conclusions.

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Data availability statement

The data associated with the study are not deposited into a publicly available repository. The data will be made available request.

CRediT authorship contribution statement

Dan Sun: Writing – review & editing, Validation, Supervision, Funding acquisition. **Fangfang Wang:** Writing – review & editing, Funding acquisition. **Qingwen Li:** Writing – review & editing. **Yisen Chen:** Writing – review & editing. **Rongjian Chen:** Supervision. **Zhichen Yang:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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